Bureau
International des
Poids et
Mesures

# Le système international d'unités
# The International System of Units

**9 édition 2019**

Annexe 2
**Appendix 2**

## Mise en pratique

## *Mise en pratique*

Partie 4.1
**Part 4.1**

## Mise en pratique de la définition de l'ampère et d'autres unités électriques

## *Mise en pratique* for the definition of the ampere and other electric units in the SI

May 20, 2019
20 mai 2019

**Bureau International**
**des Poids et Mesures**

# *Mise en pratique*
# Part 4.1: *Mise en pratique* for the definition of the ampere and other electric units in the SI

**9th edition 2019**

# 1. Introduction

The purpose of this *Mise en pratique*, prepared by the Consultative Committee for Electricity and Magnetism (CCEM) of the International Committee for Weights and Measures (CIPM), is to indicate how the SI base unit, the ampere, symbol A, and the derived electric SI units with names and symbols, the volt V, ohm Ω, siemens S, coulomb C, farad F, henry H, watt W, tesla T, and weber Wb, may be realized in practice.

In general, the term “to realize a unit” is interpreted to mean the establishment of the value and associated uncertainty of a quantity of the same kind as the unit that is consistent with the definition of the unit. The definition of the ampere does not imply any particular experiment for its practical realization. Any method capable of deriving an electric current value traceable to the set of seven reference constants could, in principle, be used. Thus, the list of methods given is not meant to be an exhaustive list of all possibilities, but rather a list of those primary methods that are easiest to implement and/or that provide the smallest uncertainties.

A primary method is a method having the highest metrological properties; whose operation can be completely described and understood; for which a complete uncertainty statement can be written down in terms of SI units; and which does not require a reference standard of the same quantity.

# 2. Definition of the ampere

The definition of the ampere, SI base unit of electric current, is as follows [1]:

**The ampere, symbol A, is the SI unit of electric current. It is defined by taking the fixed numerical value of the elementary charge $e$ to be $1.602\,176\,634 \times 10^{-19}$ when expressed in the unit C, which is equal to A s, where the second is defined in terms of $\Delta\nu_{\mathrm{Cs}}$.**

# 3. Practical realization of the ampere

As already noted in Chapter 1, to realize a unit generally means to establish the value and associated uncertainty of a quantity of the same kind as the unit that is consistent with the definition of the unit. In practice, the ampere A can be realized:

1. by using Ohm's law, the unit relation A = V/Ω, and using practical realizations of the SI derived units the volt V and the ohm Ω, based on the Josephson and quantum Hall effects, respectively, as discussed in Chapter 4 and Chapter 5 below; or
2. by using a single electron transport (SET) or similar device, the unit relation A = C/s, the value of $e$ given in the definition of the ampere and a practical realization of the SI base unit the second s; or
3. by using the relation $I = C \cdot \mathrm{d}U/\mathrm{d}t$, the unit relation A = F V/s, and practical realizations of the SI derived units the volt V and the farad F and of the SI base unit second s.

At the time of the preparation of this *Mise en pratique*, single electron transport (SET) implementations still have technical limitations and often larger relative uncertainties than some other competitive techniques. However, SET implementations are included in this *Mise en pratique* because they offer unique and elegant approaches to realizing SI units, and their uncertainties have been improving in recent years, and they promise to improve further in the future.

# 4. Practical realization of the volt, V, SI derived unit of electric potential difference (voltage) and electromotive force

The volt V can be realized using the Josephson effect and the following value of the Josephson constant $K_J$[(1)]:

[(1)] This value has been calculated to 15 significant digits.

$$K_J = 483\,597.848\,416\,984\ \mathrm{GHz\ V^{-1}} \tag{1}$$

This value follows from the assumption of the accuracy of the equation $K_J = 2e/h$, which is strongly supported by a large body of experimental and theoretical works, and the values of $h$ and $e$ given in [1]. Although the quotient $2e/h$ can obviously be calculated with any number of digits, this truncated recommended value is in error by less than 1 part in $10^{15}$, which is intended to be negligible in the vast majority of applications. In those rare cases where this error may not be negligible, additional digits should be employed. The advantage of recommending a particular value of $K_J$ for practical use is that it ensures that virtually all realizations of the volt based on the Josephson effect employ exactly the same value.

Note that the value of $K_J$ in Equation (1) is smaller than the value $K_{J\text{-}90} = 483\,597.9\ \mathrm{GHz\ V^{-1}}$, which was adopted by the CIPM starting 1 January 1990 for the international realization of the volt using the Josephson effect, by the fractional amount $106.665 \times 10^{-9}$. This implies that the unit of voltage realized using $K_{J\text{-}90}$ was larger than the present SI unit as realized using the value in Equation (1) by the same fractional amount. Thus, the numerical value of a voltage measured in terms of $K_{J\text{-}90}$ would have been smaller by the same fractional amount as the numerical value of the identical voltage measured today in terms of the present SI volt realized using the value of $K_J$ given in Equation (1).

# 5. Practical realization of the ohm, Ω, SI derived unit of electric resistance and impedance

The ohm Ω can be realized as follows:

1. by using the quantum Hall effect in a manner consistent with the CCEM Guidelines [2] and the following value of the von Klitzing constant $R_K$[(1)]:

$$R_K = 25\,812.807\,459\,304\,5\ \Omega \tag{2}$$

This value follows from the assumption of the accuracy of the equation $R_K = h/e^2$, which is strongly supported by a large body of experimental and theoretical works, and the values of $h$ and $e$ given in [1]. Although the quotient $h/e^2$ can obviously be calculated with any number of digits, this truncated recommended value is in error by less than 1 part in $10^{15}$, which is intended to be negligible in the vast majority of applications. In those rare cases where this error may not be negligible, additional digits should be employed. The advantage of recommending a particular value of $R_K$ for practical use is that it ensures that virtually all realizations of the ohm based on the quantum Hall effect employ exactly the same value; or

2. by comparing an unknown resistance to the impedance of a known capacitance using, for example, a quadrature bridge, where, for example, the capacitance has been determined by means of a calculable capacitor and the value of the electric constant given by Equation (3) in Chapter 13 below.

(1) This value has been calculated to 15 significant digits.

Note that the value of $R_K$ in Equation (2) is larger than the value $R_{K\text{-}90} = 25\,812.807\ \Omega$, which was adopted by the CIPM starting 1 January 1990 for the international realization of the ohm using the quantum Hall effect, by the fractional amount $17.793 \times 10^{-9}$. This implies that the unit of resistance realized using $R_{K\text{-}90}$ was larger than the present SI unit as realized using the value in Equation (2) by the same fractional amount. Thus, the numerical value of a resistance measured in terms of $R_{K\text{-}90}$ would have been smaller by the same fractional amount as the numerical value of the identical resistance measured today in terms of the present SI ohm realized using the value of $R_K$ given in Equation (2).

# 6. Practical realization of the siemens, S, SI derived unit of electric conductance

The siemens S can be realized from a realization of the ohm (see Chapter 5, p. 8) since S is related to Ω by the unit relation $S = \Omega^{-1}$.

# 7. Practical realization of the coulomb, C, SI derived unit of electric charge

The coulomb C can be realized as follows:

1. by measuring the duration in terms of the SI unit of time, the second s, of the flow of an electric current known in terms of the ampere realized as indicated in Chapter 3; or
2. by determining the amount of charge placed on a capacitance known in terms of the farad F realized by method Chapter 8, 1) or Chapter 8, 2), using the unit relation $C = F \cdot V$ and by measuring the voltage across the capacitance in terms of the volt V as realized by the Josephson effect and the value of the Josephson constant given in Equation (1) (see Chapter 4, p. 7); or
3. by using a SET or similar device to transfer a known amount of charge based on the value of *e*, given in the definition of the ampere, onto a suitable circuit element.

# 8. Practical realization of the farad, F, SI derived unit of capacitance

The farad F can be realized as follows:

1. by comparing the impedance of a known resistance obtained using the quantum Hall effect and the value of the von Klitzing constant given in Equation (2) (see Chapter 5, see p. 8), including a quantized Hall resistance itself, to the impedance of an unknown capacitance using, for example, a quadrature bridge; or
2. by using a calculable capacitor and the value of the electric constant given by Equation (3).

# 9. Practical realization of the henry, H, SI derived unit of inductance

The henry H can be realized as follows:

1. by comparing the impedance of an unknown inductance to the impedance of a known capacitance with the aid of known resistances using, for example, a Maxwell-Wien bridge, where the known capacitance and resistances have been determined, for example, from the quantum Hall effect and the value of $R_K$ given in Equation (2) (see Chapter 5 and Chapter 8, see p. 11); or
2. by using a calculable inductor of, for example, the Campbell type of mutual inductor and the value of the magnetic constant $\mu_0$ given by Equation (6).

# 10. Practical realization of the watt, W, SI derived unit of power

The watt W can be realized using electrical units by using the fact that electric power is equal to current times voltage, the unit relation based on Ohm's law, $W = V^2/\Omega$, and realizations of the volt and ohm using the Josephson and quantum Hall effects and the values of the Josephson and von Klitzing constants given in Equation (1) and Equation (2) (see Chapter 4 and Chapter 5, p. 8).

# 11. Practical realization of the tesla, T, SI derived unit of magnetic flux density

The tesla T can be realized as follows:

1. by using a solenoid, Helmholtz coil or other configuration of conductors of known dimensions carrying an electric current determined in terms of the ampere realized as discussed in Chapter 3, and the value of the magnetic constant $\mu_0$ given in Equation (6) in the calculation of the magnetic flux density generated by the current carrying conductors; or
2. by using nuclear magnetic resonance (NMR) with a sample of known gyromagnetic ratio, for example, a spherical sample of pure $H_2O$ at 25 °C and the most recent recommended value of the shielded gyromagnetic ratio of the proton $\gamma_p$, given by CODATA.

# 12. Practical realization of the weber, Wb, SI derived unit of magnetic flux

The weber Wb can be realized from the tesla based on the unit relation $\mathrm{Wb} = \mathrm{T\ m^2}$ or from the volt based on the unit relation $\mathrm{Wb} = \mathrm{V\ s}$. Use can also be made of the fact that the magnetic flux quantum $\phi_0$, which characterizes the magnetic properties of superconductors, is related to $h$ and $e$ as given in [1] by the exact relation $\phi_0 = h/2e$.

# 13. Magnetic constant $\mu_0$ and related quantities

The new definitions of the kilogram, ampere, kelvin, and mole do not alter the relationships among the magnetic constant (permeability of vacuum) $\mu_0$, electric constant (permittivity of vacuum) $\varepsilon_0$, characteristic impedance of vacuum $Z_0$, admittance of vacuum $Y_0$, and speed of light in vacuum $c$. Moreover, they do not change the exact value of $c$, which is explicit in the definition of the SI base unit of length, the metre, m. The relationships among these constants are

$$\varepsilon_0 = 1/\mu_0 c^2 \tag{3}$$

$$Z_0 = \mu_0 c = \left(\mu_0/\varepsilon_0\right)^{1/2} \tag{4}$$

$$Y_0 = 1/\mu_0 c = \left(\varepsilon_0/\mu_0\right)^{1/2} = 1/Z_0 \tag{5}$$

However, the new definitions do affect the value of $\mu_0$, and hence the values of $\varepsilon_0$, $Z_0$, and $Y_0$. In particular, $\mu_0$ no longer has the exact value $4\pi \times 10^{-7}$ N A$^{-2}$ and must be determined experimentally. The value of $\mu_0$ can be obtained with a relative standard uncertainty, $u_r$, identical to that of the fine structure constant $\alpha$ from the exact relation

$$\mu_0 = \alpha \frac{2h}{ce^2} \tag{6}$$

Since $h$, $c$, and $e$ have fixed numerical values, it follows from Equation (3)-Equation (6) that

$$u_r(Y_0) = u_r(Z_0) = u_r(\varepsilon_0) = u_r(\mu_0) = u_r(\alpha) \tag{7}$$

The recommended values of $h$, $e$, $k$, and $N_A$ resulting from the 2017 CODATA special least-squares adjustment of the values of the fundamental constants [3] were the basis of the exact values used for these four constants in the new definitions of the kilogram, ampere, kelvin, and mole adopted by the 26th CGPM [1]. The 2017 special adjustment but with $h$, $e$, $k$, and $N_A$ taken to have the exact values used in the new definitions, yields the following currently recommended value of the magnetic constant:

$$\mu_0 = 4\pi\left[1 + 2.0(2.3) \times 10^{-10}\right] \times 10^{-7}\ \mathrm{N\,A^{-2}} = 12.566\,370\,616\,9(29) \times 10^{-7}\ \mathrm{N\,A^{-2}} \tag{8}$$

However, users should always compute the value from the most recent CODATA adjustment [4]. The values and uncertainties of the electric constant, characteristic impedance of vacuum and characteristic admittance of vacuum may always be obtained from the relationships of Equation (3)-Equation (6).

It should be recognized that the recommended values for $\mu_0$, $\varepsilon_0$, $Z_0$, and $Y_0$ are expected to change slightly from one future CODATA adjustment to the next, as new data that influence the value of $\alpha$ become available. Users of this document should, therefore, always employ the most up-to-date CODATA recommended values for these constants in their calculations. Of course, the values of $h$, $e$, $k$, and $N_A$ fixed by the new definitions will be unchanged from one adjustment to the next.

# References

[1] BIPM, The International System of Units (SI Brochure) [9th edition, 2019], https://www.bipm.org/en/publications/si-brochure/.

[2] F. Delahaye and B. Jeckelmann, 'Revised technical guidelines for reliable dc measurements of the quantized Hall resistance', *Metrologia*, **40(5)**, 217-223 (2003).

[3] D. Newell, F. Cabiati, J. Fischer, K. Fujii, S. G. Karshenboim, H. S. Margolis, E. de Mirandes, P.J. Mohr, F. Nez, K. Pachucki, T. J. Quinn, B. N. Taylor, M. Wang, B. Wood and Z. Zhang, 'The CODATA 2017 Values of $h$, $e$, $k$, and $N_A$ for the Revision of the SI', *Metrologia*, **55(1)** L13-L16 (2018).

[4] 'CODATA recommended values of the fundamental physical constants: 2018', https://physics.nist.gov/cuu/Constants/index.html.